

NOVY URENGOY, Russian Federation

WMO: 233580

Lat: 66.0742N

Lon: 76.5006E

Elev: 62

StdP: 100.58

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.9	-41.2	-47.0	0.0	-43.0	-45.0	0.0	-41.2	13.3	-11.3	11.7	-13.2	1.6	30	0.857

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	28.2	17.6	25.4	16.9	22.8	15.6	19.0	25.6	17.7	23.8	16.3	21.7	4.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	11.9	21.4	15.1	10.8	20.2	13.9	10.0	18.9	54.3	25.5	50.0	23.7	46.1	21.2	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.7	8.7	-44.9	30.2	3.8	2.3	-47.7	31.9	-49.9	33.3	-52.0	34.6	-54.8	36.2		
(5)			-44.1	20.2	3.0	1.7	-46.3	21.4	-48.0	22.3	-49.7	23.3	-51.9	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.9	-24.1	-23.6	-15.3	-9.2	-1.8	11.1	15.7	11.0	5.7	-4.1	-16.2	-20.8	(6)
(7)		DBStd	15.57	9.67	9.07	8.37	7.70	5.79	5.82	4.86	3.58	4.31	6.36	9.17	9.85	(7)
(8)		HDD10.0	6141	1056	942	784	576	368	56	7	30	143	438	787	953	(8)
(9)		HDD18.3	8872	1314	1175	1043	826	623	226	113	228	380	697	1037	1212	(9)
(10)		CDD10.0	351	0	0	0	0	4	89	183	62	13	0	0	0	(10)
(11)		CDD18.3	40	0	0	0	0	0	8	31	1	0	0	0	0	(11)
(12)		CDH23.3	472	0	0	0	0	0	107	351	14	0	0	0	0	(12)
(13)		CDH26.7	138	0	0	0	0	0	24	113	1	0	0	0	0	(13)
(14)	Wind	WSAvg	4.3	4.2	4.2	4.6	5.0	4.8	4.4	4.0	3.7	4.1	4.6	4.1	4.3	(14)
(15)	Precipitation	PrecAvg	807	62	55	62	61	59	66	59	77	62	95	78	71	(15)
(16)		PrecMax	921	106	106	105	107	106	114	124	147	120	166	144	106	(16)
(17)		PrecMin	603	28	17	7	9	24	31	13	40	29	45	33	27	(17)
(18)		PrecStd	82	22	23	21	23	19	20	26	24	21	23	26	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.9	-2.9	1.2	6.0	21.0	29.7	31.9	25.1	21.0	10.2	1.8	-1.2	(19)
(20)			MCWB	-3.5	-3.3	0.1	2.5	13.8	17.5	19.7	16.8	14.1	8.3	1.2	-1.5	(20)
(21)		2%	DB	-5.8	-5.1	-0.3	3.9	14.1	26.2	29.8	22.2	17.1	7.1	0.7	-3.1	(21)
(22)			MCWB	-6.3	-5.5	-1.0	1.4	9.7	16.4	18.8	16.0	12.6	6.2	0.4	-3.5	(22)
(23)		5%	DB	-8.0	-8.1	-1.8	2.2	9.1	23.9	27.7	19.7	14.4	5.1	-0.9	-4.7	(23)
(24)			MCWB	-8.6	-8.5	-2.4	0.7	6.0	15.4	17.7	14.3	11.3	4.3	-1.2	-5.1	(24)
(25)		10%	DB	-10.8	-10.6	-3.6	1.2	5.2	21.1	24.8	17.3	12.2	3.3	-3.0	-7.0	(25)
(26)			MCWB	-11.1	-10.9	-4.3	0.2	3.0	14.0	17.2	13.3	9.7	2.6	-3.5	-7.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.3	-3.4	0.2	3.0	13.5	19.6	21.1	18.2	15.5	8.6	1.2	-1.6	(27)
(28)			MCDWB	-2.9	-3.2	1.0	4.9	20.1	25.6	28.4	22.3	19.5	9.6	1.4	-1.4	(28)
(29)		2%	WB	-6.3	-5.6	-1.0	1.7	10.0	18.0	19.9	16.6	13.3	6.3	0.2	-3.5	(29)
(30)			MCDWB	-5.9	-5.2	-0.4	3.1	13.3	24.1	27.4	20.7	15.9	6.9	0.4	-3.3	(30)
(31)		5%	WB	-8.4	-8.5	-2.3	0.9	6.1	16.4	18.6	15.2	11.8	4.4	-1.2	-5.1	(31)
(32)			MCDWB	-8.0	-8.2	-1.7	1.9	8.9	22.2	25.5	18.9	14.8	5.0	-0.9	-4.7	(32)
(33)		10%	WB	-10.9	-10.9	-4.2	0.3	3.3	14.9	17.4	13.7	10.1	2.6	-3.5	-7.3	(33)
(34)			MCDWB	-10.6	-10.6	-3.6	1.3	5.0	19.9	23.6	16.6	11.9	3.2	-3.1	-7.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	8.5	9.7	9.3	6.8	9.5	9.7	8.2	6.7	5.4	7.4	7.8	(35)
(36)			MCDBR	9.5	8.8	9.4	6.6	11.6	13.7	13.2	12.0	10.1	5.8	7.4	6.2	(36)
(37)			MCWBR	9.4	8.7	9.0	5.3	7.4	6.5	5.0	6.3	6.0	4.9	7.3	6.2	(37)
(38)		5% WB	MCDWB	9.6	9.1	9.6	6.9	11.3	12.5	12.3	10.5	9.8	5.6	7.2	6.6	(38)
(39)			MCWBR	9.6	8.9	9.4	6.1	7.5	6.6	5.7	6.3	6.2	4.9	7.2	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.210	0.262	0.315	0.331	0.350	0.372	0.401	0.386	0.328	0.280	0.201	0.210	(40)	
(41)		taud	1.875	2.029	2.018	2.074	2.197	2.392	2.308	2.319	2.442	2.197	1.904	1.884	(41)	
(42)		Ebn at Noon	200	575	717	817	845	837	792	763	739	559	198	20	(42)	
(43)		Edh at Noon	19	59	99	121	120	102	108	97	68	49	17	4	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.08	0.67	2.31	4.41	5.42	5.34	4.95	3.31	1.87	0.63	0.17	0.01	(44)	
(45)		RadStd	0.01	0.07	0.16	0.38	0.35	0.49	0.59	0.41	0.25	0.07	0.02	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page